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United States Patent	12403757
Kind Code	B2
Date of Patent	September 02, 2025
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Battery pack support device for vehicle

Abstract

A vehicle battery pack support device for a battery pack mounted below a pair of side rails extending in a vehicle longitudinal direction and has a pair of side faces each facing in the longitudinal direction, includes a battery bracket for housing the battery pack and a frame bracket for coupling the side rails and the battery bracket. The battery bracket includes a first bracket covering the side faces of the battery pack, at least a portion of a top face that is continuous with the side faces, and at least a portion of a bottom face that is continuous with the side faces, the first bracket having a dimension in the vehicle width direction which is equal to or larger than the long side of the battery pack and a second bracket covering an end portion of the first bracket in the width direction coupled to the frame bracket.

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Appl. No.:	17/780780
Filed (or PCT Filed):	October 09, 2020
PCT No.:	PCT/JP2020/038377
PCT Pub. No.:	WO2021/106391
PCT Pub. Date:	June 03, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220410686 A1	Dec. 29, 2022

Foreign Application Priority Data

Publication Classification

Int. Cl.: **B60K1/04** (20190101); **B62D21/02** (20060101); **H01M50/244** (20210101);
H01M50/249 (20210101)

U.S. Cl.:

CPC **B60K1/04** (20130101); **B62D21/02** (20130101); **H01M50/244** (20210101);
H01M50/249 (20210101); B60K2001/0438 (20130101); H01M2220/20 (20130101)

Field of Classification Search

CPC: B60K (2001/0438); H01M (50/249); B62D (21/02)

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a vehicle battery pack support device.

BACKGROUND ART

(2) In recent years, from the viewpoint of an environmental impact, electric vehicles are under development also in the field of commercial vehicles such as trucks (Patent Document 1). It is desirable that a battery pack mounted in such an electric truck is arranged between frames in terms of securing side collision safety of the battery pack.

CITATION LIST

Patent Document

(3) PATENT DOCUMENT 1: Japanese Unexamined Patent Publication No. 2016-113063

SUMMARY OF THE INVENTION

Technical Problem

(4) However, there are various sizes of battery packs for running a vehicle and some of the battery packs are too large to fit between a pair of frames of the electric truck. In a case of employing such a battery pack, it is not possible to arrange the battery pack between frames.

(5) In addition, there is a demand to employ a common battery pack used in passenger cars for electric trucks in order to reduce costs. However, in case of passenger cars, the battery pack is mounted inside a vehicle body, but in case of electric trucks, the battery pack is arranged so as to be exposed to the outside.

(6) Under these circumstances, the present invention was made to provide a vehicle battery pack support device capable of securing torsional rigidity and side collision safety of a battery pack when a battery pack having a dimension that does not fit between frames of an electric truck is mounted.

Solution to the Problem

(7) The present invention was made to overcome at least some of the problems described above, and can be implemented as the following embodiments or application examples.

(8) (1) A vehicle battery pack support device according to an application example is a support

device for a battery pack that is mounted below a pair of side rails extending in a longitudinal direction of a vehicle and has a pair of side faces each facing in the longitudinal direction of the vehicle, the side faces each having a long side that is orthogonal to the longitudinal direction of the vehicle and extends beyond the side rails in a vehicle width direction, the vehicle battery pack support device comprising: a battery bracket for housing the battery pack; and a frame bracket for coupling the side rails and the battery bracket, the battery bracket including a first bracket covering the side faces of the battery pack, at least a portion of a top face that is continuous with the side faces, and at least a portion of a bottom face that is continuous with the side faces, the first bracket having a dimension in the vehicle width direction which is equal to or larger than the long side of the battery pack, and a second bracket covering an end portion of the first bracket in the vehicle width direction and coupled to the frame bracket.

(9) As described above, the vehicle battery pack support device has a structure in which the first brackets are longer than the long sides of the front side face and the rear side face of the battery pack, and cover the front side face and the rear side face to prevent an impact from being directly transferred to the battery pack even when the impact is applied from a lateral side in the vehicle width direction.

(10) In addition, the second brackets cover end portions of the first brackets in the vehicle width direction, so that it is possible to improve torsional rigidity of the battery bracket and substantially prevent fragments or the like from directly reaching the battery pack even when an object collides from a lateral side in the vehicle width direction.

(11) The battery bracket including the first brackets and the second brackets can be easily formed so as to fit the shape of the battery pack. Further, the battery bracket is coupled to the side rails via the frame bracket to support the battery pack, so that the battery pack that has a size larger than the distance between the frames of the electric truck can be easily mounted, and that torsional rigidity and side collision safety of the battery pack can be secured.

(12) (2) In the vehicle battery pack support device according to the application example of the above configuration (1), the first bracket may include: a first web that covers the side face of the battery pack facing a front side of the vehicle; a first top flange that is continuous with the first web and covers the top face; and a first bottom flange that is continuous with the first web and covers the bottom face. It is thus possible to cushion an impact applied from the front side of the vehicle, and improve front collision safety of the battery pack.

(13) (3) In the vehicle battery pack support device according to the application example of the above configuration (1), the first bracket may have a cylindrical shape integrally covering the side faces, the top face, and the bottom face. This configuration can further improve torsional rigidity of the battery bracket.

(14) (4) In the vehicle battery pack support device according to the application example of any one of the above configurations (1) to (3), the second bracket may include: a second top flange coupled to the first top flange of the first bracket; a second bottom flange coupled to the first bottom flange of the first bracket; and a second web coupling the second top flange and the second bottom flange. Therefore, the second bracket has a structure of sandwiching and holding the first bracket, which makes it possible to improve torsional rigidity of the battery bracket.

(15) (5) In the vehicle battery pack support device according to the application example of the above configuration (4), at least one of the second top flange or the second bottom flange of the second bracket may be formed to have a larger dimension in the vehicle width direction at an end portion in the longitudinal direction of the vehicle than that of a central portion. It is thus possible to reduce weight of the battery bracket and secure torsional rigidity of the battery bracket.

(16) (6) In the vehicle battery pack support device according to the application example of the above configuration (4), a space may be formed between the second web of the second bracket and a side face of the battery pack in the vehicle width direction. It is thus possible to substantially prevent a situation in which even when the second brackets are deformed due to an impact applied

from the side of the vehicle, the second brackets directly come into contact with the battery pack and transfer the impact.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic top view illustrating an overall configuration of a vehicle including a vehicle battery pack support device according to an embodiment.
- (2) FIG. 2 is an exploded perspective view of a battery pack and a battery bracket.
- (3) FIG. 3 is a cross-sectional view taken along line A-A of FIG. 1.
- (4) FIG. 4 is a schematic top view illustrating an overall configuration of a vehicle including a vehicle battery pack support device according to another embodiment.

DESCRIPTION OF EMBODIMENTS

(5) An embodiment of the present invention will be described in detail below with reference to the drawings. The drawings are not necessarily to scale. FIG. 1 is a schematic top view illustrating an overall configuration of a vehicle including a vehicle battery pack support device according to the present embodiment. The shapes of components described below are, basically, horizontally symmetrical with respect to the center of the vehicle **1** in a vehicle width direction (Y direction in the drawings). Therefore, basically, only one of left and right components is described in the following description, but the other components which are symmetrical to their counterparts have the same shapes as their counterparts.

(6) The vehicle **1** includes a motor (electric motor) (not shown) as a driving power source, and is, for example, a so-called electric vehicle. In the present embodiment in particular, the vehicle **1** is a truck vehicle provided with a cab **4** and a cargo box **6**. The cab **4** is mounted on a chassis frame, and is adapted for a driver to board. The cargo box **6** is disposed behind the cab **4** and is configured to be capable of carrying the cargo. In the drawing, the cab **4** and the cargo box **6** are outlined with a dotted line.

(7) The vehicle **1** may also be a hybrid car including an engine in addition to a motor as the driving power source. Further, the vehicle **1** is not limited to the truck vehicle, and may be any other commercial vehicle including a battery for driving the vehicle.

(8) The chassis frame is what is called a ladder frame including a pair of side rails **3L** and **3R** extending in a longitudinal direction (X direction in the drawings) of the vehicle, and a plurality of cross members **3a**, **3b**, **3c**, and **3d** arranged between the side rails **3L** and **3R** along the vehicle width direction (Y direction in the drawings). The side rails are simply referred to as side rails without specifying left and right side rails. The chassis frame having such a configuration can achieve both of static strength for enduring a weight of the truck vehicle and dynamic strength (fatigue strength) for enduring repetitive load application caused by road surface vibrations or the like during traveling.

(9) Further, front wheels **7** are suspended at a front portion of the chassis frame and rear wheels **8** are suspended at a rear portion of the chassis frame. A driving force is transmitted to these wheels by the motor (not shown) such that the vehicle **1** can travel. A battery for feeding power to the motor as such a driving power source is mounted in the vehicle **1** via a support device **10**.

(10) The battery has a configuration in which a battery unit (not shown) including a plurality of secondary battery cells is housed in a battery pack **2** which is a casing. The battery pack **2** is supported on the side rails **3R** and **3L** constituting the chassis frame, via the support device **10** as illustrated in FIG. 1.

(11) The support device **10** includes a battery bracket **20** and a frame bracket **30**.

(12) The battery bracket **20** includes a pair of front first bracket **21A** and rear first bracket **21B** (also collectively referred to as first brackets **21A** and **21B**), and a pair of left second bracket **22L** and

right second bracket **22R** (also collectively referred to as second brackets **22L** and **22R**).

(13) A front portion and a rear portion of the second bracket **22L** of the battery bracket **20** are coupled to and supported on the side rail **3L** via the frame bracket **30**, and a front portion and a rear portion of the second bracket **22R** of the battery bracket **20** are coupled to and supported on the side rail **3R** via the frame bracket **30**.

(14) FIG. 2 is an exploded perspective view of the battery pack and the battery bracket. The battery pack **2** is formed in a rectangular parallelepiped shape and has a front side face **2a** which is a rectangular side face facing a front side of the vehicle, and a rear side face **2b** which is a rectangular shape facing a rear side of the vehicle. The battery pack **2** is arranged along the vehicle width direction (Y direction in the drawings) so that long sides of the front side face **2a** and the rear side face **2b** are orthogonal to the longitudinal direction (X direction in the drawings) of the vehicle. The battery pack **2** is dimensioned so that both end portions of the battery pack **2** in the vehicle width direction extend beyond outer side faces of the side rails **3L** and **3R** in the vehicle width direction. A length **L2** of each of the long sides of the front side face **2a** and the rear side face **2b** is larger than a length **L1** between the side rails **3L** and **3R**.

(15) An under cover **23** is a member on which the battery pack **2** is directly placed, and is implemented by, for example, a plate-shaped member. The under cover **23** is a cover that simply covers a bottom face of the battery pack **2**, and it is the first brackets **21A** and **21B** that directly hold the battery pack **2**.

(16) The front first bracket **21A** includes a first web **21Aw** for covering the front side face **2a** of the battery pack **2** in the drawing. Further, a first top flange **21At** for covering a portion of a top face which is continuous with the front side face **2a** of the battery pack **2**, and a first bottom flange **21Ab** for covering a portion of the bottom face which is continuous with the front side face **2a** are formed so as to be vertically continuous with the first web **21Aw**. In other words, the first top flange **21At** and the first bottom flange **21Ab** are arranged so that flat surface portions thereof face each other in a vehicle height direction (Z direction in the drawings). Thus, the front first bracket **21A** is formed in a substantial “U” shape having a cross section that is open toward the rear side of the vehicle.

(17) Similarly, the rear first bracket **21B** includes a first web **21Bw** for covering the rear side face **2b** of the battery pack **2**. Further, the rear first bracket **21B** includes a first top flange **21Bt** for covering a portion of the top face which is continuous with the rear side face **2b** of the battery pack **2**, and a first bottom flange **21Bb** for covering a portion of the bottom face which is continuous with the rear side face **2b**. In other words, the first top flange **21Bt** and the first bottom flange **21Bb** of the rear first bracket **21B** are arranged so that flat surface portions thereof face each other in the vehicle height direction (Z direction in the drawings). Thus, the rear first bracket **21B** is formed in a substantial “U” shape having a cross section that is open toward the front side of the vehicle.

(18) The open side of the first bracket **21A** and the open side of the first bracket **21B** face each other, and the first brackets **21A** and **21B** support the battery pack **2** while sandwiching the battery pack **2** therebetween in the longitudinal direction of the vehicle.

(19) The left second bracket **22L** includes a second web **22Lw** covering open faces of left end portions of the first brackets **21A** and **21B** in the vehicle width direction. Further, the second web **22Lw** has, at an upper end thereof, a second top flange **22Lt** which extends across the first top flanges **21At** and **21Bt** included in the first brackets **21A** and **21B**, respectively, and which is coupled to outer sides of the first top flanges **21At** and **21Bt** of the first brackets **21A** and **21B**. Similarly, the second web **22Lw** has, at a lower end thereof, a second bottom flange **22Lb** which extends across the first bottom flanges **21Ab** and **21Bb** included in the first brackets **21A** and **21B**, respectively, and which is coupled to outer sides of the first bottom flanges **21Ab** and **21Bb** of the first brackets **21A** and **21B**.

(20) The second top flange **22Lt** and the second bottom flange **22Lb** of the left second bracket **22L** are each formed to have a larger dimension in the vehicle width direction at an end portion in the

longitudinal direction of the vehicle than that of a central portion. Specifically, a central portion of each of the second top flange **22Lt** and the second bottom flange **22Lb** in the longitudinal direction of the vehicle is cut out in an arc shape. By doing so, torsional rigidity of a battery pack support device is improved while achieving weight reduction.

(21) The right second bracket **22R** is horizontally symmetrical to the left second bracket **22L** and has a similar configuration, and thus a description thereof will be omitted.

(22) In the drawing, when portions connected by a dot-and-dash line are aligned with each other, the battery bracket **20** surrounds four side faces and portions of the top face and the bottom face of the battery pack **2**, like a box. The battery bracket **20** surrounding the battery pack **2** as described above is coupled to the side rails **3L** and **3R** via the frame bracket **30**.

(23) Referring to FIG. **3**, FIG. **3** is a cross-sectional view taken along line A-A of FIG. **1**.

(24) Hereinafter, a left side portion of the support device **10** suspended on the side rail **3L** will be described by way of example with reference to the drawing. A right side portion of the support device **10** suspended on the side rail **3R** is horizontally symmetrical to the left side portion of the support device **10** and has a similar configuration, and thus a detailed description thereof will be omitted.

(25) As described above, the battery pack **2** is placed on the under cover **23**, and the front portion and the rear portion of the battery pack **2** are covered by the first brackets **21A** and **21B**, respectively. Specifically, the first webs **21Aw** and **21Bw** and the first top flanges **21At** and **21St** of the first brackets **21A** and **21B** are in contact with corresponding faces of the battery pack **2** (the front side face **2a**, the rear side face **2b**, and a portion of the top face), respectively, and the first bottom flanges **21Ab** and **21Bb** are in contact with the under cover **23**.

(26) In addition, the left second bracket **22L** covers the under cover **23** and the first brackets **21A** and **21B** from the outside while leaving a space **S** between the left second bracket **22L** and a left side face of the battery pack **2** in the vehicle width direction. The size of the space **S** can be adjusted by changing dimensions of the second top flange **22Lt** and the second bottom flange **22Lb** of the left second bracket **22L** in the vehicle width direction, and can be adjusted also by changing dimensions of the first brackets **21A** and **21B** in the vehicle width direction.

(27) The second bottom flange **22Lb** of the left second bracket **22L** is in contact with the first bottom flanges **21Ab** and **21Bb** of the first brackets **21A** and **21B**. The second bottom flange **22Lb** of the left second bracket **22L**, the first bottom flanges **21Ab** and **21Bb** of the first brackets **21A** and **21B**, and the under cover **23** are coupled to one another with a plurality of bolts **40** (only one bolt is illustrated in FIG. **3**).

(28) The second top flange **22Lt** of the left second bracket **22L** is in contact with the first top flanges **21At** and **21Bt** of the first brackets **21A** and **21B**, and the frame bracket **30**. The second top flange **22Lt** of the left second bracket **22L**, the first top flanges **21At** and **21Bt** of the first brackets **21A** and **21B**, and a lower portion **30c** of the frame bracket **30** are coupled to one another with a plurality of bolts **41** (only one bolt is illustrated in FIG. **3**).

(29) The frame bracket **30** is, for example, a member including an upper portion **30a**, a central portion **30b**, and the lower portion **30c**, and having an “S” shaped cross section as illustrated in the drawing. A rib (not shown) is provided on a side face of the frame bracket **30** to increase strength. The upper portion **30a** of the frame bracket **30** is coupled to a mount **31** with a plurality of bolts **42** (only one bolt is illustrated in FIG. **3**).

(30) The mount **31** is a member connecting the frame bracket **30** and the side rail **3L** to each other. For example, the mount **31** is coupled to a web **3Lw** of the side rail **3L** with a plurality of bolts **43** (only two bolts are illustrated in FIG. **3**). The mount **31** is, for example, a rubber mount, and includes an elastic material such as rubber therein to elastically hold the frame bracket **30** with respect to the side rail **3L**.

(31) As described above, the support device **10** has a structure in which the first brackets **21A** and **21B** having rigidity substantially equivalent to the chassis frame are longer than the long sides of

the front side face **2a** and the rear side face **2b** of the battery pack **2**, and cover the front side face **2a** and the rear side face **2b** to prevent an impact from being directly transferred to the battery pack **2** even when the impact is applied from a lateral side in the vehicle width direction.

(32) In addition, the second brackets **22L** and **22R** cover end portions of the first brackets **21A** and **21B** in the vehicle width direction, so that it is possible to improve torsional rigidity of the battery bracket **20** and substantially prevent fragments or the like from directly reaching the battery pack **2** even when an object collides from a lateral side in the vehicle width direction.

(33) The battery bracket **20** including the first brackets **21A** and **21B** and the second brackets **22L** and **22R** can be easily formed so as to fit the shape of the battery pack **2**. Further, the battery bracket **20** is coupled to the side rails **3L** and **3R** via the frame bracket **30** to support the battery pack **2**, so that the battery pack **2** that has a size larger than the distance between the side rails **3L** and **3R** of the electric truck can be easily mounted, and that torsional rigidity and side collision safety of the battery pack **2** can be secured.

(34) Particularly, in the front first bracket **21A**, the first top flange **21At** and the first bottom flange **21Ab** are integrally and continuously formed with the first web **21Aw** covering the front side face **2a** of the battery pack **2**, so that it is possible to cushion an impact applied from the front side of the vehicle, and improve front collision safety of the battery pack.

(35) Further, the second brackets **22L** and **22R** include the second top flanges **22Lt** and **22Rt** coupled to the first top flanges **21At** and **21Bt**, and the second bottom flanges **22Lb** and **22Rb** coupled to the first bottom flanges **21Ab** and **21Bb** of the first brackets **21A** and **21B**, and thus have a structure of sandwiching and holding the first brackets **21A** and **21B**. As a result, it is possible to improve torsional rigidity of the battery bracket **20**.

(36) In addition, the second top flanges **22Lt** and **22Rt** and the second bottom flanges **22Lb** and **22Rb** of the second brackets **22L** and **22R** are each formed to have a larger dimension in the vehicle width direction at an end portion in the longitudinal direction of the vehicle than that of a central portion. It is thus possible to reduce weight of the battery bracket **20** and secure torsional rigidity of the battery bracket **20**.

(37) Further, the space **S** is formed between the second webs **22Lw** and **22Rw** of the second brackets **22L** and **22R**, and side faces of the battery pack **2** in the vehicle width direction. It is thus possible to substantially prevent a situation in which even when the second brackets **22L** and **22R** are deformed due to an impact applied from the side of the vehicle, the second brackets **22L** and **22R** directly come into contact with the battery pack **2** and transfer the impact.

(38) Although the description of the embodiment of the vehicle battery pack support device of the present invention ends here, the present invention is not limited to this embodiment.

(39) For example, FIG. **4** is a schematic top view of a vehicle including a vehicle battery pack support device according to another embodiment. According to this embodiment, as illustrated in FIG. **4**, three battery packs **2A**, **2B**, and **2C** are arranged in a longitudinal direction of the vehicle and supported by support devices **10A**, **10B**, and **10C**, respectively.

(40) The battery pack **2B** and the support device **10B** arranged at the center in the longitudinal direction of the vehicle are similar to those of the embodiment described above, whereas the battery pack **2A** and the support device **10A** arranged at a front side of the vehicle each have a longer dimension in the vehicle width direction and are offset toward the right side of the vehicle. The battery pack **2C** and the support device **10C** arranged at a rear side of the vehicle each have a longer dimension in the longitudinal direction of the vehicle than those of the central battery pack **2B** and the support device **10B**.

(41) The support devices **10A**, **10B**, and **10C** can easily fit the shapes or layouts of the battery packs **2A**, **2B**, and **2C**, respectively. It is thus possible to obtain similar effects to those of the embodiment described above.

(42) Further, the front and rear first brackets **21A** and **21B** of the battery bracket **20** are paired according to the embodiment described above, but are not limited thereto. For example, two

bracket members each having an “L” shaped cross section may be vertically combined to function as one of the first brackets. Alternatively, the pair of front and rear first brackets **21A** and **21B** may be integrally formed in a cylindrical shape. This configuration further improves torsional rigidity of the battery bracket **20**.

(43) Although the embodiments of the present invention have been described above, the present invention can be implemented in various other forms, and various omissions, replacements, and changes can be made without departing from the spirit of the present invention. The embodiments and variations are included in the scope and spirit of the invention, as well as in the invention of claims and the equivalents thereof.

DESCRIPTION OF REFERENCE CHARACTERS

(44) **1** Vehicle **2**, **2A**, **2B**, **2C** Battery Pack **3L**, **3R** Side Rail **10**, **10A**, **10B**, **10C** Support Device **20** Battery Bracket **21A**, **21B** First Bracket **22L**, **22R** Second Bracket **23** Under Cover **30** Frame Bracket **31** Mount

Claims

1. A vehicle battery pack support device for a battery pack that is mounted below a pair of side rails extending in a longitudinal direction of a vehicle and has a pair of side faces each facing in the longitudinal direction of the vehicle, the vehicle battery pack support device comprising: a battery bracket for housing the battery pack; and a frame bracket for coupling the side rails and the battery bracket, the battery bracket including: a first bracket covering the side faces of the battery pack, at least a portion of a top face that is continuous with the side faces, and at least a portion of a bottom face that is continuous with the side faces, the first bracket having a dimension in the vehicle width direction which is equal to or larger than the long side of the battery pack, and a second bracket covering an end portion of the first bracket in the vehicle width direction and coupled to the frame bracket; wherein the first bracket comprises a plurality of bracket members each having a U-shaped cross-section, wherein the U-shaped cross-section is defined by: a web portion that covers a corresponding side face of the battery pack; a top flange portion continuous with the web portion that covers at least a portion of the top face of the battery pack; and a bottom flange portion continuous with the web portion that covers at least a portion of the bottom face of the battery pack; wherein the plurality of bracket members are arranged along the longitudinal direction of the vehicle so as to collectively cover the side faces, the top face, and the bottom face of the battery pack.

2. The vehicle battery pack support device of claim 1, wherein the second bracket includes: a second top flange coupled to the respective top flange portions of the plurality of bracket members of the first bracket; a second bottom flange coupled to the respective bottom flange portions of the plurality of bracket members of the first bracket; and a second web coupling the second top flange and the second bottom flange.

3. A vehicle battery pack support device for a battery pack that is mounted below a pair of side rails extending in a longitudinal direction of a vehicle and has a pair of side faces each facing in the longitudinal direction of the vehicle, the vehicle battery pack support device comprising: a battery bracket for housing the battery pack; and a frame bracket for coupling the side rails and the battery bracket. the battery bracket including: a first bracket covering the side faces of the battery pack, at least a portion of a top face that is continuous with the side faces, and at least a portion of a bottom face that is continuous with the side faces, the first bracket having a dimension in the vehicle width direction which is equal to or larger than the long side of the battery pack, and a second bracket covering an end portion of the first bracket in the vehicle width direction and coupled to the frame bracket; wherein the second bracket includes: a second top flange coupled to a first top flange of the first bracket; a second bottom flange coupled to a first bottom flange of the first bracket; and a second web coupling the second top flange and the second bottom flange; wherein at least one of

the second top flange or the second bottom flange of the second bracket is formed to have a larger dimension in the vehicle width direction at an end portion in the longitudinal direction of the vehicle than that of a central portion.

4. The vehicle battery pack support device of claim 2, wherein a space is formed between the second web of the second bracket and a side face of the battery pack in the vehicle width direction.
